

Task Context

You are an expert in software configuration analysis. Your task is to read the provided code carefully and analyze whether there is a dependency relationship between the given two options.

Reasoning Instructions (with symbolic knowledge)

When given a piece of code, follow these steps:

1. Identify all code variables related to the configuration options.

2. Analyze if these configuration options influence each other based on conditional logic, function calls, and assignments.

3. If the dependency can be determined, output in the “Successful Analysis Format”.

4. If the dependency cannot be determined due to missing information, and you require the definition of some variables, functions, or classes, return a request in the “Missing Definitions Format”.

Successful Analysis Format:

Answer: "Yes" or "No",
Explanation: "Reasoning behind analysis."

Missing Definitions Format:

Answer: "Unknown",
Missing: [{"name": "Name",
"type": "function"|"class"|"variable"}]

Example 1: Code snippet: Relevant code for options O_x and O_y , $\mathbf{C}_{x,y}$

Output: Whether the two options involved have dependency

The Actual Task (with symbolic knowledge)

Now, please analyze the provided code and configuration options O_α and O_β , then output using the format demonstrated above.

Configuration option name O_α : O_α

Configuration option name O_β : O_β

Probable dependency snippet: $\mathbf{C}_{i,\alpha,\beta}$

Variable point: $\mathbf{S}_{i,\alpha,\beta}$

Task Context

You are an expert in software configuration analysis. Your task is to carefully read the provided software manual, analyze the dependency relationships between configuration options described in the documentation, and output the identified dependencies using a clearly defined BNF grammar format.

Reasoning Instructions (with symbolic knowledge)

When given a piece of documentation, follow these steps:

1. Identify the most relevant descriptions of configuration options.

2. Analyze how these configuration options influence each other based on descriptions such as default values, valid conditions, usage constraints, or fallback behaviors.

3. Express the dependency using the given BNF grammar rules.

If there is no dependency between the two configuration options, output "None"; otherwise, provide your output in the following format:

Formula: $\langle \text{Dependency} \rangle$
Explanation: "Reasoning behind analysis."

The BNF grammar you should follow: see Table 1.

Example 1: Doc snippet: Descriptions for options O_x and O_y , $\mathbf{D}_x$ and $\mathbf{D}_y$

Output: BNF of dependency between options O_x and O_y

The Actual Task (with symbolic knowledge)

Now, please analyze options O_α and O_β in the provided descriptions and output the configuration dependency using the format demonstrated above.

Configuration option name O_α : O_α

Configuration option name O_β : O_β

Description for O_α : $\mathbf{D}_\alpha$

Description for O_β : $\mathbf{D}_\beta$